BOKUN ZHENG

PHYSICS SIMULATION | 3D RECONSTRUCTION | ROBOTICS

Blacksburg, VA | +1 410-528-3197 | zhengbk@hotmail.com | linkedin.com/in/bokunz | github.com/zzzzzbk

PROFILE

Ph.D. candidate specializing in physics-based simulation and 3D reconstruction, with a strong background in robotics, computational modeling, optimization and AI. Experienced in developing computational models and perception pipelines to reconstruct, simulate, and analyze complex deformable physical systems. Interested in bridging simulation and reality through bidirectional sim–real transfer in robotics.

SKILLS

Physics & Simulation: Physics-based simulation, numerical methods, solid mechanics, contact/collision, Cosserat rod theory, finite element analysis (FEA), Gazebo, MuJoCo, Issac sim

Perception & World Modeling: 3D reconstruction, computer vision, point cloud processing, motion capture, multi-camera rigs, ultrasound imaging, SLAM, Blender, 3D rendering, 3D Gaussian Splatting, stereo vision, segmentation (SAM)

Robotics & Learning: ROS, classical control, force/torque sensing, continuum robotics, cable-driven, motion planning, Model Predictive Control (MPC), mechatronics, reinforcement learning, autoencoders (VAE), foundation models, PyTorch, TensorFlow

Engineering: Python, C++, MATLAB, Git, Bash, Arduino, mechanical design (SolidWorks, Fusion), rapid prototyping, 3D printing

EDUCATION

Virginia Tech | Ph.D., Mechanical Engineering — **Expected May 2027**

Johns Hopkins University | M.S., Mechanical Engineering — **May 2021**

Tianjin University | B.E., Processing Equipment and Control Engineering — **Jul 2018**

INDUSTRY EXPERIENCE

Research Intern | Analog Devices, Inc. — **May 2026 - Aug 2026**

Developed computational design and optimization tools for advanced 6-axis force/torque sensors based on novel MEMS-enabled strain sensing for robotic applications.

- Built an AI-assisted natural-language-to-CAD-to-FEA workflow in Python with the SolidWorks API to generate, simulate, and evaluate mechanical designs programmatically.
- Built an optimization framework linking geometry, structural response, and sensing performance for 6-axis force/torque sensors across load ranges and form factors, and used it to design a novel flexure that converts deformation into measurable strain patterns for ADI's MEMS-enabled strain sensing.
- Contributed to early-stage dexterous-hand concepts through system-level work on sensing, mechanisms, and robotic manipulation.

RESEARCH EXPERIENCE

Research Assistant | Virginia Tech — **Aug 2023 – Present**

Develop computational methods for physics-based simulation and 3D reconstruction of complex deformable systems, using octopus arms to model physical interaction, recover motion, and investigate sim-real transfer. (Advisor: Noel Naughton)

- **Physics-Based Simulation:** Developed high-dimensional Cosserat-rod model of octopus arms to simulate coupled deformation, muscle activation, contact, and hydrodynamic effects, providing a physics-based model for interpreting and reproducing observed motion.
- **Nonlinear FEA:** Developed and implemented a nonlinear, anisotropic constitutive model of octopus arm muscle with C++ plugin and python scripting in FEA (FEBio), identifying how muscle activation tunes effective bending stiffness.
- **Contact-Rich Soft Robotics:** Modeled and simulated an entanglement-based soft robotic gripper with 16 independent tentacles in Isaac Sim, using simulation to optimize morphology and topology and analyze how passive mechanics enable robust contact-rich manipulation with reduced sensing and control complexity.
- **Multi-Camera Underwater Imaging:** Designed and deployed an 8-camera synchronized underwater imaging system with refractive calibration, ray tracing, and multi-view geometry to reconstruct 3D geometry and 4D motion of deformable, self-occluding animals in vivo.
- **3D Reconstruction:** Built an end-to-end pipeline from synchronized multi-view RGB data, combining visual hull / space carving geometric priors with refractive MVS and Gaussian Splatting (2DGS) for reconstructing high deformable 3D.
- **Multimodal Sensing & Fusion:** Investigated ultrasound imaging as a complementary underwater sensing modality for vision occlusion and developed data-processing methods to integrate optical and ultrasound measurements at the point-cloud level.

Research Assistant | Virginia Tech **Nov 2021 - Aug 2023**

Applied deep learning and motion analysis to learn bat's maneuvering flight and translate biological flight principles into flapping-wing robotic design. (Advisor: Rolf Mueller)

- Built a 3D bat-flight dataset by reconstructing motion-capture sequences in Blender and extracting full-body skeletal kinematics across 43 bones.
- Developed a variational autoencoder that reduced flight-motion dimensionality by 75%, then explored the latent space for motion generation and data-driven control.
- Advised iterative mechanical development of a bat-inspired flapping-wing drone, with a 20% weight-reduction objective guiding design tradeoffs.

Research Assistant | Johns Hopkins University **Jan 2020 - Aug 2021**

Developed physics-based numerical models to study bio-inspired locomotion through cluttered and obstacle-dense environments. (Advisor: Chen Li)

- Built a physics-based simulator from scratch with rigid-body dynamics, collision/contact detection, and mechanical constraints to study locomotion through cluttered environments.
- Applied Markov Chain Monte Carlo methods to model stochastic locomotion and showed how random body perturbations can facilitate obstacle traversal; published in IEEE Robotics and Automation Letters.

SELECTED PROJECT EXPERIENCE

- **Open-Source Numerical Methods:** Contributed to PyElastica, an open-source Cosserat-rod simulation framework, extending numerical tools for modeling slender and highly deformable structures.
- **Physics-Grounded Octopus Simulator:** Developed a full-body octopus simulator with muscle-actuated Cosserat-rod models for all eight arms, and baked the resulting physics-based deformation into Blender to produce realistic underwater motion and whole-body animation.
- **PyElastica GUI & MuJoCo Integration:** Developed an interactive desktop interface and MuJoCo bridge for simulation setup, environment interaction, data loading, playback, and analysis.
- **Humanoid Locomotion via DRL:** Trained a MuJoCo biped to walk an average of 900 steps while benchmarking 10+ state-of-the-art deep reinforcement learning algorithms.
- **High-DoF Snake Kinematics & Actuation:** Reconstructed 1,200+ DoF motion-capture kinematics and used MPC to estimate muscle actuation and environmental interaction inputs.
- **Mobile Robot Autonomy:** Created ROS/C++/Gazebo packages for TurtleBot teleoperation, SLAM, and autonomous navigation.
- **Mobile Robot with SLAM and Autonomous Navigation:** Developed custom ROS/C++/Gazebo packages for TurtleBot simulation, enabling teleoperation, SLAM, and autonomous navigation.
- **Model Predictive Control on Quadrotor Drones:** Implemented a robust stabilization control for quadrotor using nonlinear dynamics and real-time MPC, optimized performance through hyperparameter tuning.

SELECTED PUBLICATIONS

- Boron A*, Zheng B*, Zhou Z, Naughton N, Li S. **Stochastic entanglement of deterministic origami tentacles for robust robotic grasping.** *Advanced Science (2026).* *Equal contribution.
- Zheng B, Xuan Q, Li C. **A minimalistic stochastic dynamics model of cluttered obstacle traversal.** *IEEE Robotics and Automation Letters, 7(2), 5119-5126 (2022).*
- Zheng B, Naughton N. **Fiber-reinforced mechanical intelligence simplifies octopus arm reaching control.** *Manuscript prepared for submission (2026).*

SELECTED PRESENTATIONS

- Ultrasound-incorporated tracking of octopus arm 3D kinematics – Virginia Robotics Symposium 2026, poster presenter, **best poster award** (2026).
- Octopus arm stiffening through active exploitation of passive fibrous tissue response - Society for Integrative and Comparative Biology, talk presenter (2026).
- Testing whether potential energy landscape can predict stochastic obstacle traversal - *American Physical Society*, talk presenter (2021)